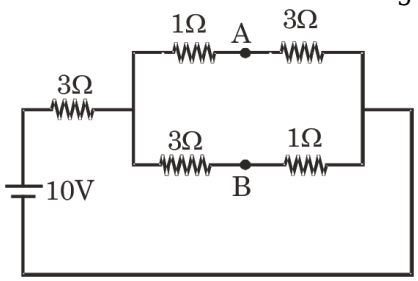
PART 1 : PHYSICS

SECTION-I (i)

1) A battery of emf 10V is connected to resistance as shown in figure. The potential difference V_A – V_B between the points A and B is :-



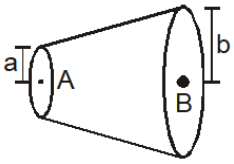
(A) 2V

(B) –2V

(C) 5V

(D) $\frac{20}{11}$V

2) An electric current passes through non uniform cross-section wire made of homogeneous and isotropic material. If the j_A and j_B be the current densities and E_A and E_B be the electric field intensities at A and B respectively, then



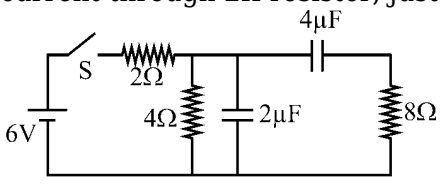
(A) $j_A > j_B$; $E_A > E_B$

(B) $j_A > j_B$; $E_A < E_B$

(C) $j_A < j_B$; $E_A > E_B$

(D) $j_A < j_B$; $E_A < E_B$

3) In the circuit shown below, initiallly the switch is open and the capacitors are uncharged. The ratio of current through 2Ω resistor, just after the switch is closed and a long time after the switch is closed :



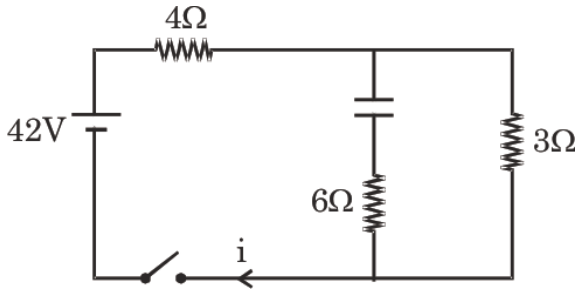
(A) 2 : 1

(B) 1 : 2

(C) 3 : 1

(D) 1 : 3

4) Initially capacitor is uncharged and at t = 0 switch is closed. The value of i after a very long time is :-



(A) 6A

(B) 5A

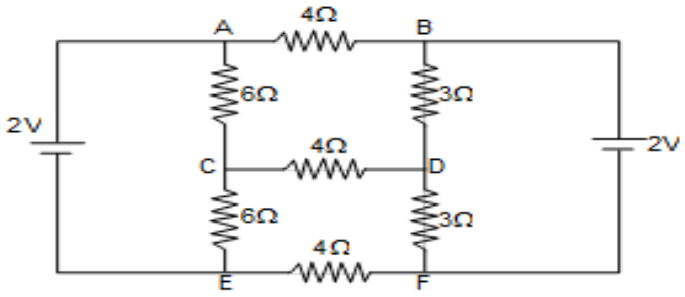
(C) 7A

(D) 4A

SECTION-I (ii)

1)

Consider following circuit and select correct alternative(s)



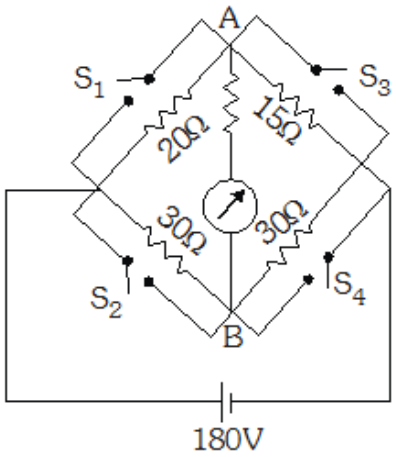
(A) Current through each cell is $\frac{1}{6}$A

(B) Current through each 6Ω resistance is $\frac{1}{6}$ A

(C) Current through each 3Ω resistance is $\frac{1}{3}$A

(D) Current through each 4Ω resistance is $\frac{1}{2}$A

2) Consider the circuit shown. The resistance connected between the junction A and B is 60 Ω including the resistance of the galvanometer. The switches have no resistance when shorted and infinite resistance when opened. All the switches are initially open and they are closed as given in



the options. Choose correct option(s) :

(A) If only switch S_1 is closed current through the battery is 20 Amp

(B) If only switch S_2 is closed then current in galvanometer from B to A

(C) If only switch S_3 is closed then current in galvanometer from B to A

(D) If only switch S_4 is closed then current in galvanometer from A to B

3) Starting at time t = 0 from the origin with speed 1 ms^{-1}, a particle follows a two-dimensional trajectory in the x-y plane so that its coordinates are related by the equation $y = \frac{x^2}{2}$. The x and y components of its acceleration are denoted by a_x and a_y, respectively. Then

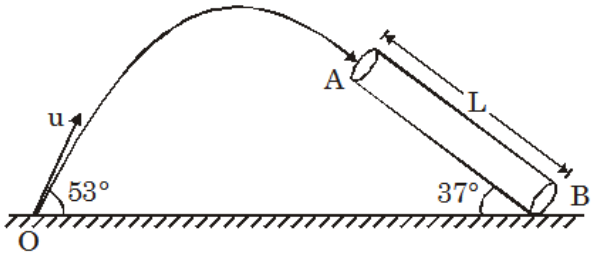
(A) $a_x = 1\ ms^{-2}$ implies that when the particle is at the origin, $a_y = 1\ ms^{-2}$

(B) $a_x = 0$ implies $a_y = 1\ ms^{-2}$ at all times

(C) at t = 0, the particle's velocity points in the □-direction

(D) $a_x = 0$ implies that at t = 1 s, the angle between the particle's velocity and the □ axis is 45°

4) AB is a pipe fixed to the ground at an inclination of 37°. At t = 0, a ball is projected from point O at a speed of u = 20 m/s at an angle of 53° to the horizontal and it smoothly enters into the frictionless pipe with its velocity parallel to the axis of the pipe. [Take : g = 10 ms^{-2}]



(A) The length 'L' of the pipe is $L = \frac{175}{12}m$

(B) The distance of end B of the pipe from point O is $\frac{125}{3}m$

(C) Speed of the particle when it reaches at the end B of the pipe is 15 m/s.

(D) Total time taken by particle to move from point 'O' to end 'B' of the pipe is 3.2 sec.

5) A projectile is thrown with a speed u at angle θ with the horizontal. The radius of the curvature

when the velocity makes an angle $\left(\frac{\theta}{2}\right)$ with the horizontal is :

(A) $\frac{u^2\cos^2\theta}{2g\cos^3(\theta/2)}$

(B) $\frac{u^2\sin^2\theta}{g\sin^3(\theta/2)}$

(C) $\frac{u^2\cos^2\theta}{g\cos^3(\theta/2)}$

(D) $\frac{2u^2\sin^2\theta}{g\sin^3(\theta/2)}$

6) A car starts from station and moves along the horizontal road by a machine delivering constant power. The distance covered by the car in time t is proportional to

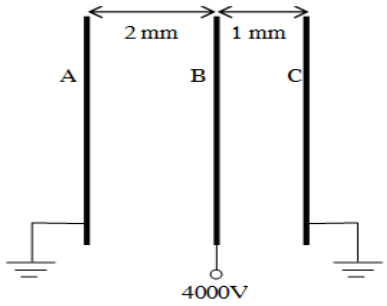
(A) t^2

(B) $t^{3/2}$

(C) $t^{2/3}$

(D) t^3

7) Three large conducting plates A,B and C each of area 2×10^{-2} m^2 are kept parallel to each other at separation shown in figure. Initially A and C are earthed and B is charged to a potential of 4000 V. All the three plates are now insulated and the plate B is removed, then



choose the correct option(s) :

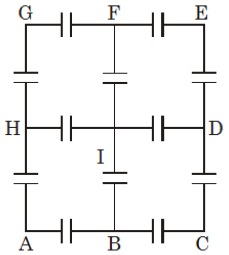
(A) The final potential difference between plates A and C is 3000 V

(B) The final potential difference between plates A and C is 0 V

(C) The change in energy stored in the region between plates A and C is approxmately = –1.86 mJ

(D) The change in energy stored in the region between plates A and C is approximately = –3.72 mJ

8) Consider capacitance circuit consisting 12 identical capacitors each of capacitance lμF as shown



in the figure. The equivalent capacitance between the terminals

(A) A and E is $\frac{2}{3}\mu F$

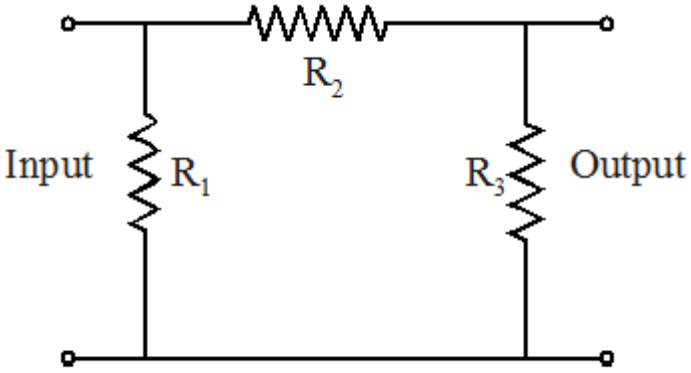
(B) A and I is $\frac{8}{7}\mu F$

(C) H and I is $\frac{12}{7}\mu F$

(D) H and B is $\frac{5}{7}\mu F$

SECTION-II

1) If at the input circuit (Figure) a voltage V_1 = 100 V, is applied the potential difference at the output V_2 = 40 V. If at the output circuit a voltage V_3 = 60 V is applied, the potential difference at the input will be V_4 = 15 V. resistor R_2 = 60 ohms. Find electrical resistance of resistors R_1 and R_3 (in



Ohm) and fill $R_1 + R_3$ in OMR sheet.

2) A galvanometer gives full scale reading of 50 mA, when a potential difference. across its terminals is 0.15 V. It can be used as an ammeter of range 0-100 A by connecting a shunt resistance of $P \times 10^{-3}$ Ω. Value of P is.

3) In the circuit shown (figure 1), all the switches are open initially and all capacitors are uncharged. Now first only S_1 is closed and then opened when steady state is reached. Then S_2 is closed and again opened when steady state is reached, after this S_3 is closed and opened when again steady state is reached, this process repeated countineously. After closing sixth switch, the fourth capacitor (8C) is removed and connected (say at t = 0) to an uncharged capacitor of original capacitance C which now has a dielectric (dielectric constant 2) filling half its volume (see figure 2), through a resistor R. If final charge on capacitor B when steady state reached in figure 2 is $\frac{8}{n}CV$. The value of n is

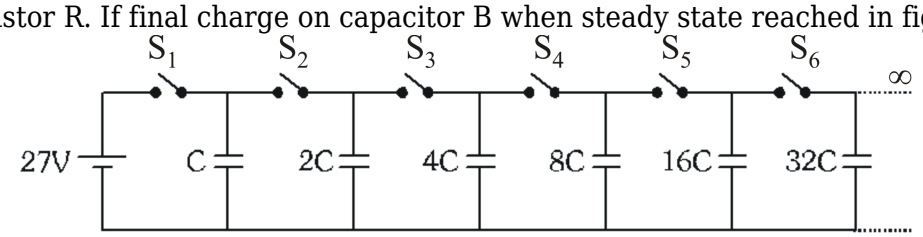


Fig.(1)

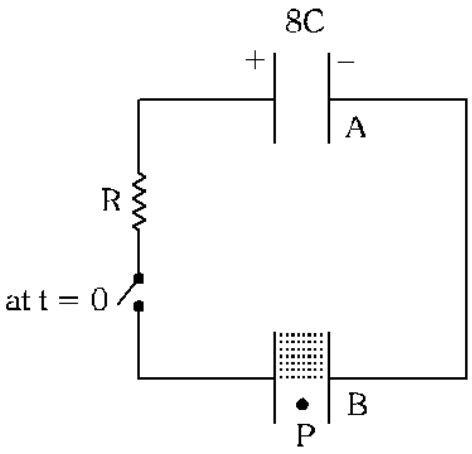
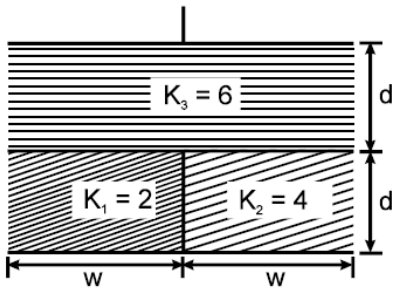


Fig. (2)

4)

A parallel plate capacitor of capacitance C (without dielectrics) is filled by dielectric slabs as shown in figure. Then the new capacitance of the capacitor is nC then n is



5) A massive ball is falling down from an initial height of h = 20 m. With a gun held horizontally, d = 50 m far from the trajectory of the falling ball, at the height of $h' = 10m$, we are going to shoot at the falling ball. The bullet leaves the gun at a speed of v = 100 m/s. At what time (in seconds) after the start of the fall should the gun be fired in order to hit the falling ball with the bullet ? (Ignore air resistance) Take ($g = 10\ m/s^2$)

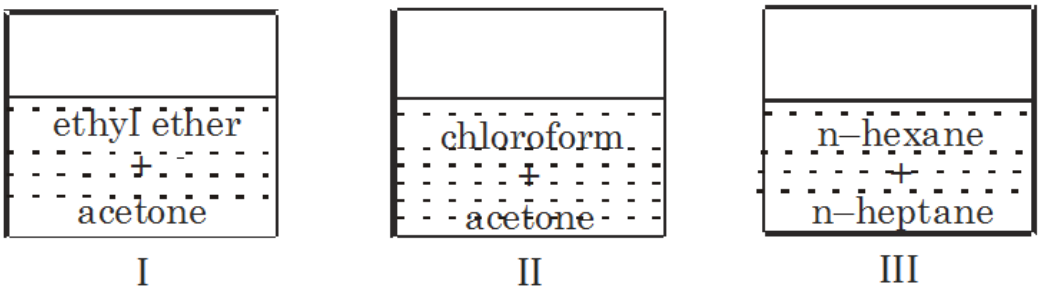
6) A projectile takes off with an initial velocity of 50 m/s at an angle of 37° with horizontal. It is just able to clear two hurdles of height 25 m each, separated from each other by a distance d (in m). Calculate d.

PART 2 : CHEMISTRY

SECTION-I (i)

1)

State T(true)/F(false) for following statements regarding given liquid solutions :



(a) Final volume of solutions if equal volume of compounds are mixed, is in the order:
$V_I = V_{II} < V_{III}$
(b) ΔS_{mix} for solution I and solution II is positive but for solution III it is zero.
(c) Enthalpy change of mixing is negative for solution II and solution III but positive for solution I.

(A) TFF
(B) FFF
(C) TTF
(D) FTT

2) A current of 0.0965 ampere is passed for 1000 seconds through 50ml of 0.1 M NaCl, using inert electrodes the average concentration of OH^- in the final solution is

(A) 0.01 M
(B) 0.02 M
(C) 0.03 M
(D) 0.04 M

3) Calculate ΔGº for

$$Mn^{+2} + MnO_2 \xrightarrow{H^+} 2Mn^{+3} + H_2O$$

Given : $E^o_{Mn^{+3}/Mn^{+2}} = 1.51\,V$, $E^o_{MnO_2/Mn^{+3}} = 0.95\,V$

(A) 237.39 kJ
(B) 54 kJ
(C) –54 kJ
(D) –237.39 kJ

4) Electromagnetic radiations having λ = 310 Å are subjected to a metal sheet having work function = 12.8 eV. What will be the velocity of photoelectrons with maximum Kinetic Energy....

(A) 0, no emission will occur
(B) 2.18×10^6 m/s
(C) $2.18 \times \sqrt{2} \times 10^6$ m/s
(D) 8.72×10^6 m/s

SECTION-I (ii)

1)

Choose the correct relation on the basis of Bohr's theory -

(A) velocity of electron $\propto \frac{1}{n}$

(B) frequency of revolution $\propto \frac{Z^2}{n^3}$

(C) radius of orbit $\propto n^2Z$

(D) force on electron $\propto \frac{Z^3}{n^4}$

2) In which of these options do both constituents of the pair have the same magnetic moment -

(A) Zn^{2+} and Cu^{+}

(B) Co^{2+} and Ni^{2+}

(C) Mn^{4+} and Co^{2+}

(D) Mg^{2+} and Sc^{+}

3) Identify the option(s) which are not correct w.r.t. solubility of a gas in water.

(A) Value of K_H (Henry's constant) will decrease as the temperature increases.

(B) Value of K_H (Henry's constant) at same temperature for O_2 gas will be less as compared to H_2 gas

(C) Soft drinks will have greater "fizz" if sealed under low pressure.

(D) Scuba divers when they move underwater will have increased solubility of gases in their blood.

4) Mark out the correct statement(s) regarding electrolytic molar conductivity.

(A) It increases as temperature increases.

(B) It experiences resistance due to vibration of ion at the mean position.

(C) Increase in concentration decreases the electrolytic molar conductivity of both the strong as well as the weak electrolyte.

(D) Greater the polarity of solvent, greater is the electrolytic molar conduction.

5) Which are correct match :-

(A) O > F > N > C — IInd I.P.

(B) $S^{-2} > Cl^{-} > K^{+} > Ca^{+2}$ — Ionic radius

(C) N > C > P > Si — E. N.

(D) F > Na > Ne — I^{st} I.P.

6) Select the correct statement-

(A) IE of N^{-} is less than IE of N

(B) Ionization Energy of an atom is always an endothermic process

(C) Magnitude of $(IE)_{Na}$ is always equal to $(EA)_{Na^{+}}$

(D) First $\Delta_{eg}H$ of any element is always positive

7) Which of the following are correct ?

(A) $IE_2(Mg) < IE_2(Na)$

(B) EA (N) < EA (P)

(C) Atomic size Mg^{+2} > Atomic size (Li^+)

(D) IP of Na < Mg < Al

8) When a lead storage battery is discharged then :

(A) SO_2 is evolved

(B) lead sulphate is produced at both electrodes

(C) sulphuric acid is consumed

(D) water is formed

SECTION-II

1) How many pairs are, in which first species has lower ionisation energy than second species.
(i) N and O (ii) Br and K (iii) Be and B (iv) I and I^-
(v) Li and Li^+ (vi) O and S (vii) Ba and Sr

2) The radial wave function for 1s orbit for a single electron specie is given by $\psi_r = ke^{-\sigma/2}$ where $\sigma = \frac{2Zr}{na_0}$ and K is constant [a^0 = radius of first Bohr's orbital in H-atom = 52.9 pm, n = Principal quantum number.)
Calculate the most probable distance of 1s electron in Be^{3+} ion. Give your answer in the form most probable distance in pm?

3) Calculate magnitude of standard electrode reduction potential in volts when silver wire coated with a layer of silver iodide is placed in solution of KI at 25ºC.
Given : $E^º_{Ag^+/Ag}$ = 0.8 V ; $K_{sp}(AgI) = 10^{-16}$; $\frac{2.303\ RT}{F} = 0.06$

4) The reduction potential (E°, in V) of MnO_4^-(aq)/Mn(s) is.....
[Given: $E^0_{(MnO_4^-(aq)/MnO_2(s))} = 1.68V; E^0_{(MnO_2(s)/Mn^{2+}(aq))} = 1.21V; E^0_{(Mn^{2+}(aq)/Mn(s))} = -1.03V$]

5) Elevation in boiling point for 1.5 molal solution of glucose in water is 4K. The depression in freezing point for 4.5 molal solution of glucose in water is 4K. The ratio of molal elevation constant to molal depression constant $\left(\frac{K_b}{K_f}\right)$ is _____.

6) Liquids 'A" and 'B' form an ideal solution. Calculate the mole-fraction of 'A' in the vapours above the liquid solution containing the liquids 'A' and 'B' in 2 : 3 mole ratio, at equilibrium.
[**Given :** P_A^0 = 0.4 atm, P_B^0 = 0.8 atm]

PART 3 : MATHEMATICS

SECTION-I (i)

1) The sum of series $\tan^{-1}\left(\frac{\sqrt{3}}{9}\right)+\tan^{-1}\left(\frac{\sqrt{3}}{19}\right)+\tan^{-1}\left(\frac{\sqrt{3}}{33}\right)+\tan^{-1}\left(\frac{\sqrt{3}}{51}\right)+.....$ up to infinite terms is equal to

(A) $\frac{\pi}{6}$
(B) $\frac{\pi}{3}$
(C) $\frac{\pi}{4}$
(D) $\frac{2\pi}{3}$

2) a,b,c are real numbers greater then 1 and $x \in R^+$. If $\log_a x = 24$, $\log_b x = 40$, $\log_{abc} x = 12$ then $\log_c x$ equal is-

(A) 80
(B) 60
(C) 48
(D) 72

3) The number of values of x satisfying the inequality $\log_{(x^2+2)}(x^2-4x+6) \geq \log_{(|\sin x|+1)}(x^2-4x+6)$ is -

(A) 0
(B) 1
(C) 3
(D) 5

4) The number of solution(s) of the equation $\log_{\frac{1}{3+|x|}}\left(5+x^2\right)=\log_{3+x^2}\left(17+x^{\frac{1}{4}}\right)$

(A) 1
(B) 2
(C) 3
(D) 0

SECTION-I (ii)

1) $\sum_{r=1}^{\infty} \tan^{-1}\left(\frac{2}{(r+2)^2}\right)$ is-

(A) $\pi - \left(\tan^{-1}2 + \tan^{-1}3\right)$

(B) $\frac{\pi}{4}$

(C) $\frac{7\pi}{4}$

(D) $\pi + \left(\tan^{-1}2 + \tan^{-1}3\right)$

2) If $2^{\left(\cos^{-1}x\right)^2} \cdot \sqrt{\left(\tan^{-1}y\right)^4 - 2\left(\tan^{-1}y\right)^2 + 2} = 1$ then the value of (x – y) can be

(A) $1 - \frac{\pi}{4}$

(B) $1 + \frac{\pi}{4}$

(C) 1 + tan 1

(D) 1 – tan 1

3) If $f(x) = \sin^{-1}\left(x^2 + \frac{1}{x^2} - 1\right) + \cos^{-1}(\sec^2(x^2 - 1))$, then correct option/s is/are

(A) Number of integral values of x in the domain of f is 2

(B) f is a constant function

(C) Number of integral values of x in the domain of f is infinite

(D) f is an even function

4) For any positive integer n, let $S_n : (0, \infty) \to \mathbb{R}$ be defined by $S_n(x) = \sum_{k=1}^{n} \cot^{-1}\left(\frac{1 + k(k+1)x^2}{x}\right)$, where for any $x \in \mathbb{R}$, $\cot^{-1}x \in (0, \pi)$ and $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then which of the following statements is (are) TRUE ?

(A) $S_{10}(x) = \frac{\pi}{2} - \tan^{-1}\left(\frac{1 + 11x^2}{10x}\right)$, for all $x > 0$

(B) $\lim_{n\to\infty} \cot(S_n(x)) = x$, for all $x > 0$

(C) The equation $S_3(x) = \frac{\pi}{4}$ has a root in $(0, \infty)$

(D) $\tan(S_n(x)) \leq \frac{1}{2}$, for all $n \geq 1$ and $x > 0$

5) Consider $f(x) = \sin^{-1}[2x] + \cos^{-1}([x] - 1)$ (where [.] denotes greatest integer function.) If domain of

$f(x)$ is $[a,b)$ & the range of $f(x)$ is $\{c,d\}$ then $a + b + \frac{2d}{c}$ is divisible by (where $c < d$)

(A) 4

(B) 2

(C) 8

(D) 9

6) Let $f : R \to (0, \pi)$ be defined as $f(x) = \cot^{-1}\left(\frac{|x|-2}{|x|+2}\right)$. If range of $f(x)$ is $(a, b]$, then -

(A) $a + b = \pi$

(B) $a + b = \frac{3\pi}{4}$

(C) $\frac{a}{b} = 3$

(D) $\frac{a}{b} = \frac{1}{3}$

7) Which of the following is/are correct ?

(A) $2^{\sqrt{\log_3 2}} = 3^{\sqrt{\log_2 3}}$

(B) If $\log_{12}27 = x$, then $\log_6 16 = 4\left(\frac{3-x}{3+x}\right)$

(C) $2^{\sqrt{\log_2 3}} = 3^{\sqrt{\log_3 2}}$

(D) $\log_3 5 > \log_{19} 25$

8) If $\log_2 3 \log_3 4 \log_4 5 \ldots\ldots \log_n(n + 1) = 10$ and $\log_{10} 2 = 0.3010$, then which of the following is true

(A) Characteristic of $(n + 1)$ to the base 32 is 2

(B) Number of zeroes after decimal and before first significant digit in $\left(\frac{1}{n+1}\right)^4$ is 12

(C) If $n = (p_1)(p_2)(p_3)$ where p_1, p_2, p_3 are prime numbers, then $p_1 + p_2 + p_3 = 45$.

(D) n is a prime number.

SECTION-II

1) Find the value of λ for which $\tan^{-1}\left\{\frac{\cos 2\alpha \sec 2\beta + \cos 2\beta \sec 2\alpha}{\lambda}\right\} = \tan^{-1}\left\{\tan^2(\alpha+\beta)\tan^2(\alpha-\beta)\right\} + \tan^{-1} 1$

2) The value of $\cos^{-1}\left[\cot\left\{\sin^{-1}\left(\sqrt{\frac{2-\sqrt{3}}{4}}\right) + \cos^{-1}\left(\frac{\sqrt{12}}{4}\right) + \sec^{-1}\left(\sqrt{2}\right)\right\}\right]$ is equal to

3) If $S = \left\{x \in \mathbb{R} : \sin^{-1}\left(\frac{x+1}{\sqrt{x^2+2x+2}}\right) - \sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right) = \frac{\pi}{4}\right\}$, then $\sum_{x \in R}\left(\sin\left(\left(x^2+x+5\right)\frac{\pi}{2}\right) - \cos\left(\left(x^2+x+5\right)\pi\right)\right)$ is equal to _____.

4) If $A = \frac{1}{1}\cot^{-1}\left(\frac{1}{1}\right) + \frac{1}{2}\cot^{-1}\left(\frac{1}{2}\right) + \frac{1}{3}\cot^{-1}\left(\frac{1}{3}\right)$ and $B = 1\cot^{-1}(1) + 2\cot^{-1}(2) + 3\cot^{-1}(3)$ then (B – A) is equal to $\left(\frac{a\pi}{b} + \frac{c}{d}\cot^{-1}(3)\right)$, where a,b,c,d ∈ N and are in their lowest form, find $\frac{(a+b+c+d)}{5}$

5) $\frac{24}{\log_2(2020)^6} + \frac{12}{\log_5(2020)^6} + \frac{12}{\log_{101}(2020)^6}$ is equal to

6) $\frac{81^{\frac{1}{\log_5 9}} + 3^{\frac{3}{\log_{\sqrt{6}} 3}}}{409} \cdot \left(\left(\sqrt{7}\right)^{\frac{2}{\log_{25} 7}} - (125)^{\log_{25} 6}\right) =$ _____.

ANSWER KEYS

PART 1 : PHYSICS

SECTION-I (i)

Q.	1	2	3	4
A.	A	A	C	A

SECTION-I (ii)

Q.	5	6	7	8	9	10	11	12
A.	B,C	B,C,D	A,B,C,D	A,B	C	B	A,C	A,B,C

SECTION-II

Q.	13	14	15	16	17	18
A.	60.00	1.50	19.00	3.90	1.00	160.00

PART 2 : CHEMISTRY

SECTION-I (i)

Q.	19	20	21	22
A.	B	B	B	C

SECTION-I (ii)

Q.	23	24	25	26	27	28	29	30
A.	A,B,D	A,C	A,C	A,C,D	A,B,C	A,B,C	A,B	B,C,D

SECTION-II

Q.	31	32	33	34	35	36
A.	2.00	13.21 to 13.23	0.16	0.77	3.00	0.25

PART 3 : MATHEMATICS

SECTION-I (i)

Q.	37	38	39	40
A.	A	B	A	D

SECTION-I (ii)

Q.	41	42	43	44	45	46	47	48
A.	A,B	C,D	A,B,D	A,B	A,B	A,D	B,C,D	A,B,C

SECTION-II

Q.	49	50	51	52	53	54
A.	**2.00**	**1.57**	**4.00**	**8.00**	**2.00**	**1.00**

SOLUTIONS

PART 1 : PHYSICS

1) $i = \frac{10}{3+2} = 2A$

$i_1 = 1A$

$i_3 = 1A$

potential difference = 3 × 1 – 1 × 1 = 2v

2)

$j = \frac{I}{A}$ and $j = \frac{E}{\rho}$

$\Rightarrow j_A > j_B$ and $E_A > E_B$.

3)

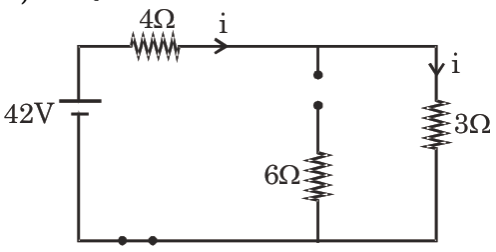
At t = 0

$I_1 = \frac{6}{2} = 3A$

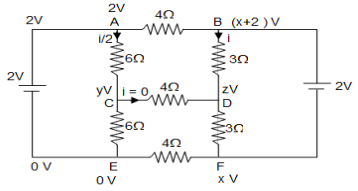
At t = ∞, $I_2 = \frac{6}{6} = 1A$

$\frac{I_1}{I_2} = \frac{3}{1}$

4) $i = \frac{42}{7} = 6A$



5)



Apply junction rule at B, C and D

$\frac{2-x-2}{4} + \frac{z-x-2}{3} = \frac{x-z}{3} + \frac{x-0}{4}$ $\frac{x}{4} + \frac{z-x}{3}$ $\frac{2}{3} = \frac{x-2}{3} + \frac{x}{4}$

4x – 4z + 3x = – 4 7x – 4z = – 4

$\left(\frac{2-y}{6} = \frac{y-z}{4} + \frac{y-0}{6}\right)$ 4 – 2y = 3y – 3z + 2y

$\left(\frac{y-z}{4}+\frac{x+2-z}{3}=\frac{z-x}{3}\right)$

$7y - 3z = 4$

$3y - 3z + 4x + 8 - 4z = 4z - 4x$

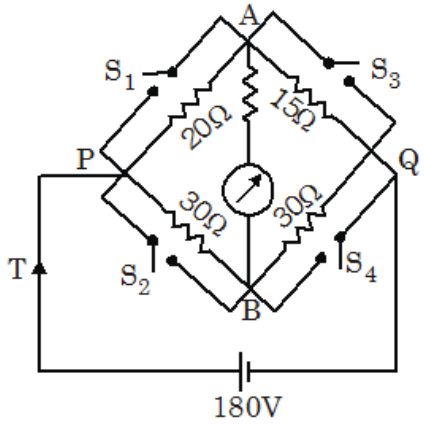
$8x + 3y - 11z = -8$

Solving Above equation we will get $x = 0, y = 1, z = 1$

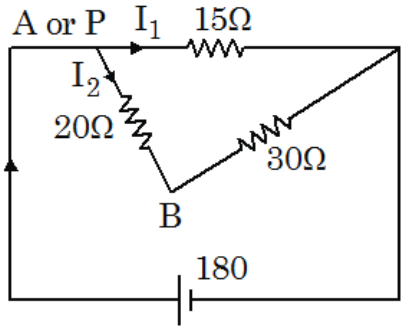
Due to symmetrical circuit, potential across 4Ω is zero and current flow will be as shown in the figure.

Hence current through 6Ω is $=\frac{2}{12}=\frac{1}{6}A$ and current through 3Ω is $=\frac{2}{6}=\frac{1}{3}A$

6) If S_1 closed. PB and AB in parallel



So effective circuit as shown



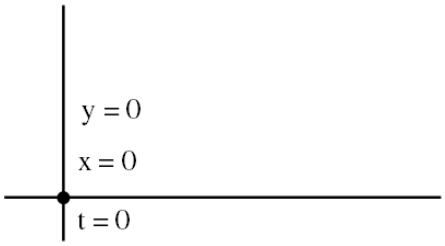
$I_1=\frac{180}{15}=12\,\text{amp}$; $I_2=\frac{180}{50}=3.6\,\text{amp}$

So I = 15.6 amp.

When S_2 closed B is at higher potential than A so current in galvanometer from B to A.

When S_3 closed B is at higher potential than A so current in galvanometer from B to A.

When S_2 closed A is at higher potential than B so current in galvanometer from A to B



7)

$$y = \frac{x^2}{2}$$

at t = 0, $\left.\begin{matrix} x = 0, y = 0 \\ u = 1 \end{matrix}\right\}$ given

$$y = \frac{x^2}{2}$$

$$\frac{dy}{dt} = \frac{1}{2}.2x\frac{dx}{dt}$$

$\Rightarrow v_y = xv_x$

difference wrt time

$$a_y = \frac{dx}{dt}.V_x + xa_x$$

$a_y = v_x^2 + xa_x$

Option

(A) If $a_x = 1$ and particle is at origin

(x = 0, y =0)

$a_y = v_x^2$

$a_y = 1^2 = 1$

At origin, at t = 0 sec

speed = 1 given

(B) Option

$a_y = v_x^2 + xa_x$

given in option B, $a_x = 0$

$\Rightarrow a_y = v_x^2$

If $a_x = 0$, v_x = constant = 1, (all the time)

$\Rightarrow a_y = I^2 = 1$ (all the time)

(C) at t = 0, x = 0 $v_y = xv_x$

speed = 1

$v_y = 0$

$v_x = 1$

(D) $a_y = v_x^2 + xa_x$

$v_y = xv_x$

$a_x = 0$ (given in D option)

$\Rightarrow a_y = v_x^2$

If $a_x = 0 \Rightarrow V_x$ = constant initially ($v_x = 1$)

$\Rightarrow a_y = 1^2 = 1$

at t = 1 sec

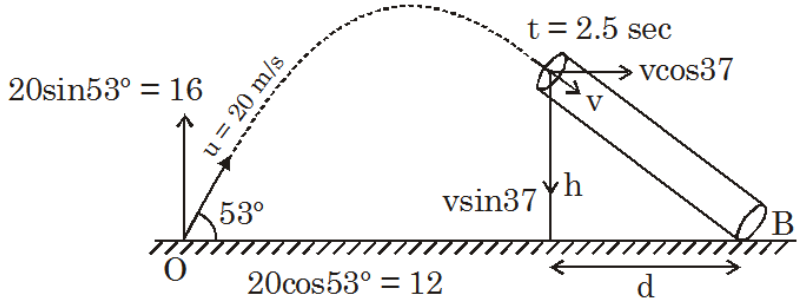
$v_y = 0 + a_y \times t = 1 \times 1 = 1$

$$\tan\theta = \frac{v_y}{v_x} = x$$

(θ → angle with x axis)

$$\tan\theta = \frac{v_y}{v_x} = \frac{1}{1} = 1$$

$\theta = 45°$



8)

$v\cos 37 = 20\cos 53$

$v = 20 \times \frac{3}{4} = 15\,m/s$

$v_y = u_y + a_y t$

$-v\sin 37 = 16 - 10t$

$-15 \times \frac{3}{5} = 16 - 10t$

$t = 2.5\ sec$

$y = u_y t + \frac{1}{2} a_y t^2$

$h = 16 \times 2.5 + \frac{1}{2} \times (-10) \times 2.5^2$

$= 40 - \frac{62.5}{2}$

$= 40 - 31.25$

$h = 8.75\,m$

$\sin 37 = \frac{h}{L}$

$L = \frac{h}{\sin 37} = \frac{8.75}{\frac{3}{5}} = \frac{875 \times 5}{300} = \frac{175}{12} m$

$x = u_x t$

$= 12 \times 2.5$

$= 30m$

$d = L\cos 37$

$d = \frac{175}{12} \times \frac{4}{5}$

$= \frac{35}{3}$

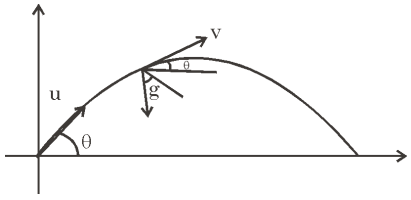
$OB = 30 + \frac{35}{3} = \frac{125}{3} m$

$L = vt + \frac{1}{2} g\sin 37° t^2$

$\frac{175}{12} = 15t + 3t^2$

$36t^2 + 180t - 175 = 0$

$t_{OB} = 2.5 + t$



9)

$v\cos\frac{\theta}{2} = u\cos\theta$

$v = \frac{u\cos\theta}{\cos\theta/2}$

$a_n = g\cos\theta/2$

$$R = \frac{v^2}{a_n} = \frac{u^2\cos^2\theta}{\frac{\cos^2\theta/2}{g\cos\theta/2}}$$

10) Power, $P = \frac{W}{t}$

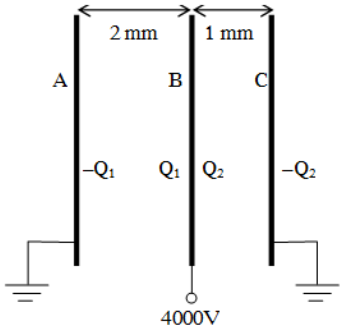
$$P = \frac{Fs}{t}, P = \frac{mas}{t} (\because F = ma)$$

$$P = \frac{mv\,s}{t^2}, \left(\because a = \frac{v}{t}\right)$$

$$P = \frac{ms}{t^3}, \left(\because v = \frac{s}{t}\right)$$

$$Pt^3 = ms^2$$

$\therefore s \propto t^{3/2}$



11)

$$Q_1 = \frac{\in_0 A}{2d}V \text{ , } Q_2 = \frac{\in_0 A}{d}V$$

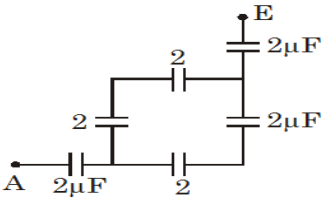
$$V_{AC} = \frac{(Q_2 - Q_1)3d}{2\in_0 A} = \frac{\in_0 AV}{2d}\cdot\frac{3d}{2\in_0 A} = \frac{3V}{4} = 3000\text{ V}$$

$$\Delta U = \frac{1}{2}\in_0\left(\frac{3V}{4.3d}\right)^2 A3d - \left[\frac{1}{2}\in_0\left(\frac{V}{2d}\right)^2 + \frac{1}{2}\in_0\left(\frac{V}{d}\right)^2 Ad\right]$$

$$= \frac{3}{32}\frac{\in_0 A}{d}V^2 - \left[\frac{3}{4}\frac{\in_0 A}{d}V\right] = -\frac{3\times 7}{32}\frac{\in_0 A}{d}V^2 = \frac{-3\times 7}{32}\times\frac{8.85\times 10^{-12}\times 2\times 10^{-2}}{10^{-3}}\times(4000)^2$$

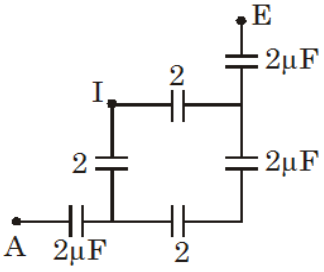
12)

(A)



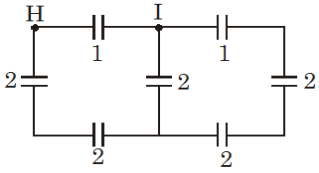
$$C_{AE} = \frac{2}{3}\mu F$$

(B)



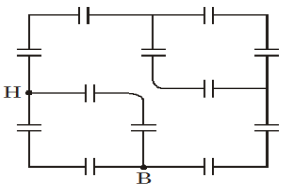
$$C_{AI} = \frac{\frac{8}{3}\times 2}{\frac{8}{3}+2} = \frac{16}{14} = \frac{8}{7}\mu F$$

(C)

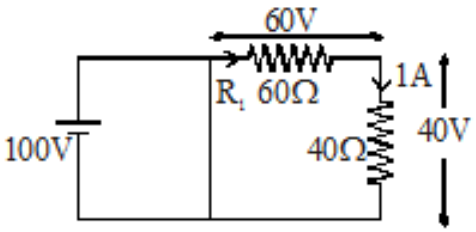


$$C_{HI} = 1 + \frac{2.5}{3.5} = \frac{12}{7}\mu F$$

(D)

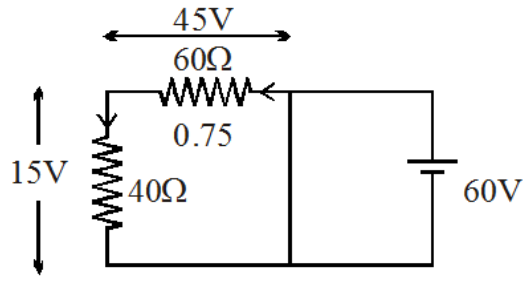


$$C_{HB} = \frac{7}{5}\mu F$$



13)

$R_3 = 40\ \Omega$



$$R_1 = \frac{15}{3/4} = 20\Omega$$

$$\frac{R_1 + R_3}{10} = 6$$

14) Galvanometer resistance

$$G = \frac{0.15V}{50mA} = \frac{150mV}{50mA} = 3\Omega$$

Full scale meter current = 50 mA =0.05 A

A galvanometer may be coverted into an ammeter of range 0 - 100 A by connecting a low resistance S parallel to the galvanometer.

$$S = \frac{I_g G}{I - I_g} = \frac{0.05 \times 3}{100 - 0.05} = \frac{0.15}{99.95} = 0.0015\Omega$$

15) After closing S_1 charge on C = 27 CV

After closing S_2 charge on

$$2C = \frac{2C \cdot 27CV}{3C} = 18\ CV$$

After closing S_3 charge on

$$4C = 4C \cdot \frac{18CV}{6C} = 12CV$$

After closing S_4 charge on

$$8C = 8C \cdot \frac{12CV}{12C} = 8CV$$

After closing S_5 charge on

$$8C = 8C \cdot \frac{8CV}{24C} = \frac{8CV}{3}$$

Final charge on 8C in figure (2)

$$Q = 8C \cdot \frac{\frac{8CV}{3}}{8C + \frac{3C}{2}} = \frac{\frac{64CV}{3}}{\frac{19}{2}} = \frac{128CV}{57}$$

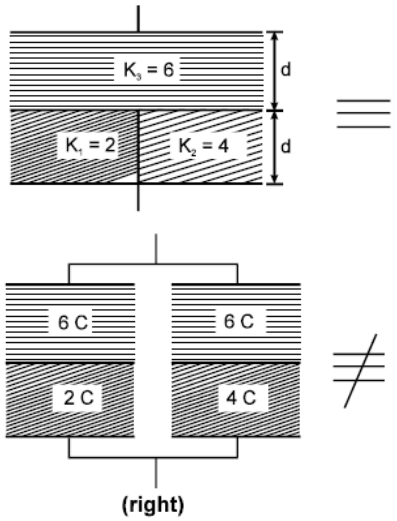
Final charge on capacitor B is

$$Q = \frac{3C}{2} \cdot \frac{\frac{8CV}{3}}{8C + \frac{3C}{2}} = \frac{8CV}{19}$$

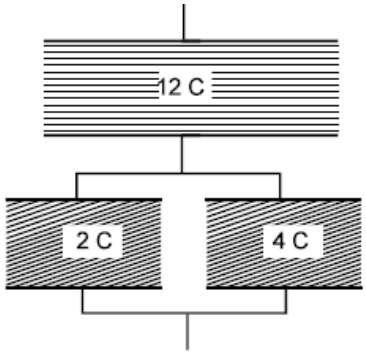
16)

The equivalent capacitance

$$C_{eq} = \frac{2C \times 6C}{2C + 6C} + \frac{6C \times 4C}{6C + 4C} = 3.9C$$



(right)



as potential difference across dielectrics of dielectric constant 2 and 4 is not same.

(wrong)

17) Let τ = delay

$y_{bullet} = y_{ball}$ (w.r.t bottom)

$$\therefore 10 + \frac{1}{2}gt^2 = \frac{1}{2}g(t+\tau)^2 \quad ...(1)$$

for 't'

d = 50 m, V = 100 m/s

d = vt

$$t = \frac{1}{2}\text{sec} \quad .... (2)$$

using (1) and (2)

$$10 + \frac{1}{2} \times 10 \times \frac{1}{4} = \frac{1}{2} \times 10\left(\frac{1}{2} + \tau\right)^2$$

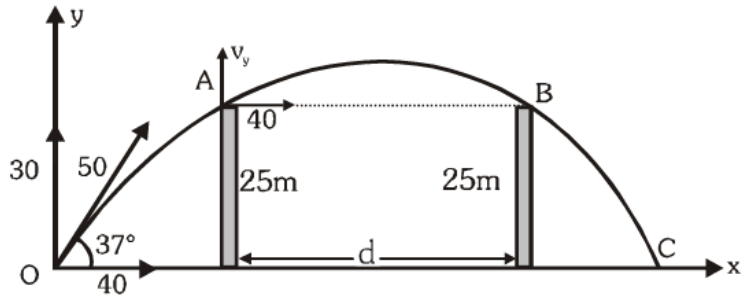
$$10^2 + \frac{10^2}{8^4} = 5\left(\frac{1}{2} + \tau\right)^2$$

$$\frac{9}{4} = \left(\frac{1}{2} + \tau\right)^2$$

$$\frac{1}{2} + \tau = \frac{3}{2}$$

$$\boxed{\tau = 1 \text{ sec}}$$

18)



$$v_y^2 = u_y^2 + 2a_y y = 30^2 + 2(-10)(25) = 900 - 500 = 400 \Rightarrow v_y = 20\text{m/s}$$

Time from A to B = $\frac{2v_y}{g} = \frac{2(20)}{10} = 4\text{s} \Rightarrow d = 40 \times 4 = 160\text{m}$

PART 2 : CHEMISTRY

19)

I. Ethyl ether + acetone : + ve deviation from Raoult's law
II. Chloroform + acetone : –ve deviation from Raoult's law
III. n–Hexane + h–heptane: ideal solution

20) At cathode : $2H_2O\ (l) + 2e^- \rightarrow H_2(g) + 2OH^-\ (aq)$

$n_{(OH^-)}F = Q \Rightarrow n_{OH^-} = \frac{Q}{96500} = \frac{(0.0965)\ (1000)}{(96500)} = \frac{1}{1000}$

$[OH^-] = \frac{\frac{1}{1000} \times 1000}{50} = \frac{1}{50} = .02\ M$

21)

$Mn^{+3} + e^- \rightarrow Mn^{+2}$; E° = 1.51 V ...(1)
$Mn^{+4} + e^- \rightarrow Mn^{+3}$; E° = 0.95 V ...(2)

$E^{º}_{cell}$ = –0.56 V
$\Delta G^{º}$ = + 1 × 96500 × 0.56
= +54 kJ

22)

$E = \frac{hc}{\lambda} \Rightarrow \frac{1240}{\lambda_{nm}} ev$ $E = \frac{1240}{31} \Rightarrow 40$ eV
40 = 12.8 + K.E.
K.E. = 40 –12.8 = 27.2 eV
K.E. = 27.2 × 1.6 × 10^{-19}
$27.2 \times 1.6 \times 10^{-19} = \frac{1}{2} \times 9.1 \times 10^{-31} \times v^2$
$v = 2.18 \times \sqrt{2} \times 10^6$ m/s

23)

(A) $v = 2.18 \times 10^6 \times \frac{Z}{n} \Rightarrow v \propto \frac{Z}{n}$ or $v \propto \frac{1}{n}$
(B) $f = \frac{v}{2\pi r}$ or $f \propto \frac{v}{r} \propto \frac{Z/n}{n^2/Z}$, $f \propto \frac{Z^2}{n^3}$
(C) $r \propto n^2 / Z$,
(D) $F = \frac{mv^2}{r}$, $F \propto \frac{v^2}{r} \propto \frac{(Z^2/n^2)}{n^2/Z}$, $F \propto \frac{Z^3}{n^4}$

24)

Species having the same number of unpaired electrons have same magnetic moment.
Spin only magnetic moment $\vec{\mu}_S = \sqrt{n(n+2)}$B.M.
where n = no. of unpaired e^-.

25) (A) On increasing temperature K_H will increase
(B) Solubility of O_2 will be higher than H_2 at same temperature
(C) If sealed under low pressure, less CO_2 will evolve due to low solubility.
(D) Under water due to high pressure solubility of gases will also be high.

26) Resistance of cell is not due to vibrations of ion but actually it is due to collisons of ions.

27)

I^{st} I.P. of Ne > F > Na

28) First $\Delta_{eg}H$ of any element is endothermic for group-2, group-18 & Nitrogen

29)

r ⇒ $Li^+ > Mg^{+2}$
IE ⇒ Na < Mg > Al

30)

Anode : Pb(s)
Cathode : PbO_2(s)
H_2SO_4(conc.) about 38% solution of H_2SO_4 is taken.
Anode : $Pb(s) \rightarrow Pb^{2+}(aq) + 2e^-$
$Pb^{2+}(aq) + SO_4^{2-}(aq) \rightarrow PbSO_4(s)$
$Pb(s) + SO_4^{2-}(aq) \rightarrow PbSO_4 + 2e^-$
most of the $PbSO_4$(s) ppt sticks to the lead rod.
Cathode : $2e^- + 4H^+ + PbO_2(s) \rightarrow Pb^{2+}(aq) + 2H_2O(l)$
$Pb^{2+}(aq) + SO_4^{2-}(aq) + 4H^+ + 2e^- \rightarrow PbSO_4(s) + 2H_2O(l)$
$PbSO_4$(s) sticks to cathode rod.
Over all reaction: $Pb(s) + PbO_2(s) + 2H_2SO_4(aq) \rightarrow 2PbSO_4(s) + 2H_2O(l)$, $E_{cell} = 2.05$ V

31) I.E. ∝ stability of half & fully filled electronic configuration
(v) & (vii)

32)

Probability of finding electron at a distance r and (r + dr), $f(r) = 4\pi r^2 \psi^2(r) = k' r^2 e - \frac{2zr}{na_0}$
For 1S orbital, n = 1
$f(r) = k' r^2 e^{-\frac{2zr}{na_0}}$
For maximum of f(r), $\frac{df(r)}{dr} = 0$

$$\Rightarrow e^{-\frac{2zr}{a_0}} \times 2r + r^2 \times e^{-\frac{2zr}{a_0}} \times \left(-\frac{2z}{a_0}\right) = 0$$

$$2 - \frac{2rz}{a_0} = 0$$

$$r = \frac{a_0}{z} = \frac{52.9}{4} = 13.225$$

33) $E^o_{I^-/AgI/Ag} = E^o_{Ag^+/Ag} + \frac{0.06}{1} \log K_{sp}(AgI)$

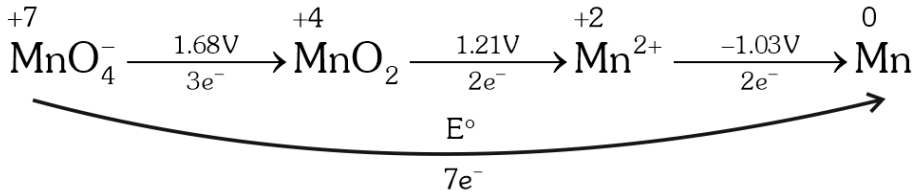
$$= 0.8 + \frac{0.06}{1} \log 10^{-16}$$

= 0.8 – 0.06 × 16 = 0.8 – 0.96 = –0.16 Volt

34) $Mn^{+7} \rightarrow Mn^{o}$

Topic : Latimer diagram

Level : Moderate



$1.68 \times 3 + 1.21 \times 2 - 1.03 \times 2 = 7E^{\circ}$

$E^{\circ} = \frac{5.4}{7} \quad \Rightarrow E^{\circ} = 0.7714V$

35) $\Delta T_b = K_b \times m_1$

$\Delta T_f = K_f \times m_2$

$\frac{\Delta T_b}{\Delta T_f} = \frac{m_1\ K_b}{m_2\ K_f}$

$\frac{K_b}{K_f} = \frac{\Delta T_b}{\Delta T_f} \times \frac{m_2}{m_1}$

$\frac{K_b}{K_f} = \frac{4}{4} \times \frac{4.5}{1.5}$

$\frac{K_b}{K_f} = 3$

36) According to Raoult's law :

$P_s = P_A^o X_A + P_B^o X_B$; where

X_A; X_B = mole fraction of A & B in the liquid phase, respectively

P_A^o, P_B^o = vapour pressure of pure A & pure B; respectively

Ps = Vapour pressure of the binary solution of A & B

$\Rightarrow$ Ps = (0.4 atm) $\left(\frac{2}{5}\right)$ + (0.8 atm) $\left(\frac{3}{5}\right)$

= 0.64 atm

According to Dalton's & Raoult's law,

$P_A = P_A^o X_A = P_S y_A$; where

Y_A = mole fraction of A in the vapours above the solution

$\Rightarrow y_A = \frac{P_A^o X_A}{P_S} = \frac{(0.4\text{atm})\left(\frac{2}{5}\right)}{0.64\ \text{atm}} = 0.25$

PART 3 : MATHEMATICS

37) $\sum_{n=2}^{\infty} \tan^{-1}\left(\frac{\sqrt{3}}{2n^2+1}\right)$

$= \sum_{n=2}^{\infty} \tan^{-1}\left[\frac{2\sqrt{3}}{4n^2+2}\right]$

$= \sum_{n=2}^{\infty} \tan^{-1}\left[\frac{2\sqrt{3}}{3+(2n+1)(2n-1)}\right]$

$$= \sum_{n=2}^{\infty} \tan^{-1}\left[\frac{\frac{2}{\sqrt{3}}}{1+\left(\frac{2n+1}{\sqrt{3}}\right)\left(\frac{2n-1}{\sqrt{3}}\right)}\right]$$

$$= \sum_{n=2}^{\infty}\left[\tan^{-1}\left(\frac{2n+1}{\sqrt{3}}\right)-\tan^{-1}\left(\frac{2n-1}{\sqrt{3}}\right)\right]$$

$$= \frac{\pi}{2}-\tan^{-1}\left(\sqrt{3}\right)=\frac{\pi}{6}$$

38)

Here $a^{24} = x\ \ b^{40} = x\ \ (abc)^{12} = x$

$\therefore a^{24} = b^{40} = (abc)^{12} = x$

$\therefore a = x^{1/24}\ \ b = x^{1/40}$ & $abc = x^{1/12}$

....(1) ....(2) ....(3)

 By (1) & (2)

$\therefore a = x^{1/24 + 1/40} \times c = x^{1/12}$

On solving $c = x^{1/60}$

 $C^{60} = x \Rightarrow \log_c x = 60$

 Option (B) is correct

39) $$\frac{1}{\log_{(x^2-4x+6)}(x^2+2)} \geqslant \frac{1}{\log_{(x^2-4x+6)}(|\sin x|+1)}$$

$x^2 - 4x + 6$ has min value 2

$\log_{(x^2-4x+6)}(|\sin x| + 1) \geq \log_{(x^2-4x+6)}(x^2+2)$

$|\sin x| + 1 \geq x^2 + 2$

$|\sin x| \geq x^2 + 1$

Intersection is not possible.

40) $$\log_{\frac{1}{3+|x|}}(5+x^2) = \log_{(3+x^2)}(17+x^{1/4})$$

$-\log_{(3+|x|)}(5+x^2) = \log_{3+x^2}(17+x^{1/4})$

(–ve) ≠ (+ve)

41) $$T_r = \tan^{-1}\left(\frac{2}{(r^2+4r+3)+1}\right) = \tan^{-1}\left(\frac{(r+3)-(r+1)}{1+(r+3)(r+1)}\right)$$

$T_r = \tan^{-1}(r + 3) - \tan^{-1}(r + 1)$

sum = $\pi - (\tan^{-1}(2) + \tan^{-1}(3))$

$= \pi - \left(\pi + \tan^{-1}(-1)\right) = \frac{\pi}{4}$

42) We have

$$2^{\left(\cos^{-1}x\right)^2}\cdot\sqrt{\left(\left(\tan^{-1}y\right)^2-1\right)^2+1}=1$$

Now $2^{\left(\cos^{-1}x\right)^2} \geq 1$ and $\sqrt{\left(\left(\tan^{-1}y\right)^2-1\right)^2+1} \geq 1$

So, it is possible its $2^{\left(\cos^{-1}x\right)^2} = 1$ and $(\tan^{-1}y)^2 - 1)^2 = 0$

So $(\cos^{-1}x)^2 = 0$ $\tan^{-1}y = \pm 1$

$\cos^{-1}x = 0$ $y = \pm\tan 1$

$x = 1$

So $x - y = 1 - \tan 1, 1 + \tan 1$.

43) $x^2 + \frac{1}{x^2} - 1 \in [-1, 1] \Rightarrow x = \pm 1$

$\sec^2\left(x^2-1\right) \in [-1, 1] \Rightarrow x = \pm 1$

So $x = \pm 1$ lies in domain only

Now $f(x) = \sin^{-1}1 + \cos^{-1}1 = \frac{\pi}{2}$

a constant function

44) $S_n(x) = \sum_{k=1}^{n} \tan^{-1}\left(\frac{x}{1+kx(kx+x)}\right)$

$= \sum_{k=1}^{n} \tan^{-1}\left(\frac{(kx+x)-(kx)}{1+(kx+x)(kx)}\right)$

$S_n(x) = \tan^{-1}(nx + x) - \tan^{-1}x = \tan^{-1}\left(\frac{nx}{1+(n+1)x^2}\right)$

(A) $S_{10}(x) = \tan^{-1}\frac{10x}{1+11x^2} = \frac{\pi}{2} - \tan^{-1}\left(\frac{1+11x^2}{10x}\right)$ $(x > 0)$

(B) $\lim_{n\to\infty} \cot(S_n(x)) = \lim_{n\to\infty} \frac{\frac{1}{n} + \left(1+\frac{1}{n}\right)x^2}{x} = x$ $(x > 0)$

(C) $S_3(x) = \tan^{-1}\frac{3x}{1+4x^2} = \frac{\pi}{4} \Rightarrow 4x^2 - 3x + 1 = 0 \Rightarrow x \notin \mathbb{R}$

(D) $\tan(S_n(x)) = \frac{nx}{1+(n+1)x^2}$; $\forall\ n \geq 1\ ;\ x > 0$

We need to check the validity of $\frac{nx}{1+(n+1)x^2} \leq \frac{1}{2}$ " $n \geq 1\ ;\ x > 0\ ;\ n \in \mathbb{N}$

$\Rightarrow 2nx \leq (n + 1)x^2 + 1$

$\Rightarrow (n + 1)x^2 - 2nx + 1 \geq 0\ \forall\ n \geq 1\ ;\ x > 0\ ;\ n \in \mathbb{N}$

Discriminant of $y = (n + 1)x^2 - 2nx + 1$ is

$D = 4n^2 - 4(n + 1)$ and $n \in \mathbb{N}$

$D < 0$ for $n = 1$; true for $x > 0$

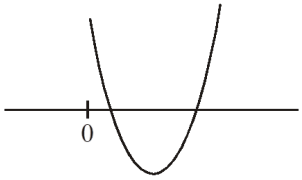
$D > 0$ for $n \geq 2 \Rightarrow \exists$ some $x > 0$

for which $y < 0$ as both roots of

$y = 0$ will be positive.

$y = (n + 1)x^2 - 2nx + 1$, $n \geq 2$

So, $y \geq 0\ \forall\ n \geq 1\ ;\ \forall\ x > 0\ ;\ n \in N$ is false.



45) $f(x) = \sin^{-1}[2x] + \cos^{-1}([x] - 1)$

$-1 \le [2x] \le 1$ & $-1 \le [x] - 1 \le 1$

$\Rightarrow -1 \le 2x < 2$ & $0 \le [x] \le 2$

$\Rightarrow -\frac{1}{2} \leqslant x < 1$ & $0 \le x < 3$

$\Rightarrow 0 \le x < 1$ Domain

$\Rightarrow [x] = 0 \Rightarrow 0 \le 2x < 2$

$\Rightarrow [2x] = 0$ or 1

Now $f(x) = \sin^{-1}[2x] + \cos^{-1}(-1)$

$= \left(0 \text{ or } \frac{\pi}{2}\right) + \pi = \pi \text{ or } \frac{3\pi}{2}$

$\Rightarrow a + b + \frac{2d}{c} = 1 + 3 = 4$

46) $y = \frac{\pi}{2} - \tan^{-1}\left(\frac{|x| - 2}{|x| + 2}\right)$

$y = \frac{\pi}{2} + \tan^{-1}\left(\frac{2 - |x|}{2 + |x|}\right) \Rightarrow y = \frac{\pi}{2} + \tan^{-1}\left(\frac{1 - \frac{|x|}{2}}{1 + \frac{|x|}{2}}\right)$

$\Rightarrow y = \frac{\pi}{2} + \left(\tan^{-1} 1 - \tan^{-1}\frac{|x|}{2}\right) \Rightarrow y = \frac{3\pi}{4} - \tan^{-1}\frac{|x|}{2}$

at $|x| = 0 \Rightarrow y = \frac{3\pi}{4}$
at $|x| \to \infty \Rightarrow y \to \frac{\pi}{4}$
$\Big\} \therefore \text{Range } \left(\frac{\pi}{4}, \frac{3\pi}{4}\right]$

47) (B) $\log_{12}27 = x = 3\log_{12}3 = \frac{3}{\log_3 12} = \frac{3}{1 + 2\log_3 2}$ $\log_3 2 = \frac{3 - x}{2x}$

$\log_6 16 = 4\log_6 2 = \frac{4}{\log_2 6} = \frac{4}{1 + \log_2 3}$

$= \frac{4}{1 + \frac{2x}{3-x}} = 4\left(\frac{3-x}{3+x}\right)$

(D) $\log_3 5 > \log_{19} 25$

$\log_3 5 > 2\log_{19} 5$

$\log_3 5 > \log_{\sqrt{19}} 5$

48) $\log_2 3 \log_3 4 \log_4 5 \ldots\ldots \log_n(n + 1) = 10$

$n + 1 = 2^{10}$

$n = 1023$

(A) $\log_{32}(n + 1) = \log_{32}1024 = 2$

(B) $\log_{10}(1024)^{-4} = -40\log_{10}2 = -12.04$

charaterstic = -13 and number of zero = 12

(C) $n = 1023 = 11.3.31$

$p_1 + p_2 + p_3 = 45$

49) RHS $= \tan^{-1}\left\{\tan^2(\alpha + \beta)\tan^2(\alpha - \beta)\right\} + \tan^{-1} 1$

$= \tan^{-1}\left\{\frac{\tan^2(\alpha + \beta)\tan^2(\alpha - \beta) + 1}{1 - \tan^2(\alpha + \beta)\tan^2(\alpha - \beta)}\right\}$

$$= \tan^{-1}\left\{\frac{\sin^2(\alpha+\beta)\sin^2(\alpha-\beta)+\cos^2(\alpha+\beta)\cos^2(\alpha-\beta)}{\cos^2(\alpha+\beta)\cos^2(\alpha-\beta)-\sin^2(\alpha+\beta)\sin^2(\alpha-\beta)}\right\}$$

$$= \tan^{-1}\left\{\frac{(\cos 2\beta-\cos 2\alpha)^2+(\cos 2\alpha+\cos 2\beta)^2}{(\cos 2\alpha+\cos 2\beta)^2-(\cos 2\beta-\cos 2\alpha)^2}\right\}$$

$$= \tan^{-1}\left\{\frac{\cos^2 2\alpha+\cos^2 2\beta}{2\cos 2\alpha\cos 2\beta}\right\}$$

$$= \tan^{-1}\left\{\frac{\cos 2\alpha\sec 2\beta+\cos 2\beta\sec 2\alpha}{2}\right\}$$

so $\lambda = 2$

50) $\sin^{-1}\left(\sqrt{\frac{2-\sqrt{3}}{4}}\right) = \sin^{-1}\left(\sqrt{\frac{4-2\sqrt{3}}{8}}\right)$

$$= \sin^{-1}\left(\sqrt{\frac{\left(\sqrt{3}-1\right)^2}{8}}\right) = \sin^{-1}\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right) = \frac{\pi}{12}$$

$$\cos^{-1}\left(\frac{\sqrt{12}}{4}\right) = \cos^{-1}\left(\frac{2\sqrt{3}}{4}\right) = \frac{\pi}{6}$$

$$\sec^{-1}\left(\sqrt{2}\right) = \frac{\pi}{4}$$

$$\cos^{-1}\left[\cot\left\{\frac{\pi}{12}+\frac{\pi}{6}+\frac{\pi}{4}\right\}\right] = \cos^{-1}\left[\cot\left\{\frac{\pi}{2}\right\}\right]$$

$= \cos^{-1}(0) = \frac{\pi}{2}$

51) $\sin^{-1}\left(\frac{(x+1)}{\sqrt{(x+1)^2+1}}\right) - \sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right) = \frac{\pi}{4}$

$\because \frac{t}{\sqrt{t^2+1}} \in (-1,1)$

$$\sin^{-1}\left(\frac{(x+1)}{\sqrt{(x+1)^2+1}}\right) = \sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right) + \frac{\pi}{4}$$

$$\frac{(x+1)}{\sqrt{(x+1)^2+1}} = \left(\frac{1}{\sqrt{2}}\right)\cos\left(\sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right)\right) + \frac{1}{\sqrt{2}}\left(\frac{x}{\sqrt{x^2+1}}\right)$$

$$= \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{x^2+1}}+\frac{x}{\sqrt{x^2+1}}\right)$$

$$\frac{(x+1)}{\sqrt{(x+1)^2+1}} = \frac{1}{\sqrt{2}}\left(\frac{1+x}{\sqrt{x^2+1}}\right)$$

After solving this equation, we get

$x = -1$ or $x = 0$

$S = \{-1, 0\}$

$$\sum_{x\in R}\left(\sin\left(\left(x^2+x+5\right)\frac{\pi}{2}\right)-\cos\left(\left(x^2+x+5\right)\pi\right)\right)$$

$= \left[\sin\left(\frac{5\pi}{2}\right) - \cos(5\pi)\right] + \left[\sin\left(\frac{5\pi}{2}\right) - \cos(5\pi)\right]$

$= 4$

52) $B - A = \left(2\cot^{-1}(2) + 3\cot^{-1}(3)\right) - \left(\frac{1}{2}\cot^{-1}\left(\frac{1}{2}\right) + \frac{1}{3}\cot^{-1}\left(\frac{1}{3}\right)\right)$

$= 2\left(\cot^{-1}2 + \cot^{-1}3\right) + \cot^{-1}3 - \left(\frac{1}{3}\left(\cot^{-1}\frac{1}{2} + \cot^{-1}\frac{1}{3}\right) + \frac{1}{6}\cot^{-1}\frac{1}{2}\right)$

$= \frac{\pi}{2} + \cot^{-1}3 - \left[\frac{\pi}{4} + \frac{1}{6}\tan^{-1}2\right] = \frac{\pi}{4} + \cot^{-1}3 - \frac{1}{6}\left[\frac{3\pi}{4} - \tan^{-1}3\right]$

$= \frac{5\pi}{24} + \frac{5}{6}\cot^{-1}3$

Hence a = 5; b = 24; c = 5; d = 6

a + b + c + d = 40

53) $4\log_{2020}2 + 2\log_{2020}5 + 2\log_{2020}101$

$= 2\log_{2020}2020 = 2$

54) $\frac{81^{\frac{1}{\log_5 9}} + 3^{\frac{3}{\log_{\sqrt{6}}3}}}{409} \cdot \left(\left(\sqrt{7}\right)^{\frac{2}{\log_{25}7}} - (125)^{\log_{25}6}\right)$

$= \frac{81^{\log_9 5} + 3^{\log_3\left(\sqrt{6}\right)^3}}{409}\left(7^{\log_7 25} - 5^{\frac{3}{2}\log_5 6}\right)$

$= \frac{\left(25 + 6^{3/2}\right)\left(25 - 6^{3/2}\right)}{409}$ $\left(\because a^{\log_a x} = x\right)$

$= \frac{625 - 6^3}{409}$ $\left(\because \log_{a^n} x = \frac{1}{n}\log x\right)$

$= \frac{409}{409}$

$= 1$